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MetalKB: Predicting Metal Binding Sites on Proteins with a Knowledge-Based Graph Framework

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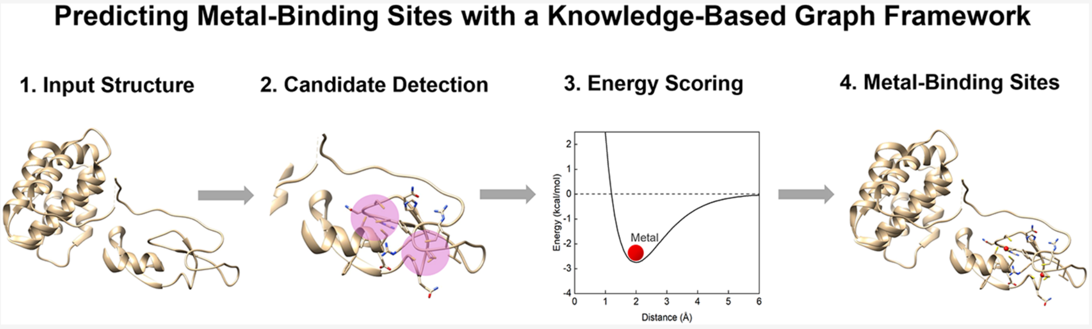

ABSTRACT: Metal ions play a crucial role in the function, regulation, and stability of proteins. Therefore, accurate prediction of metal ions' binding sites is valuable to reveal the molecular mechanism of related biological processes. Here, we propose MetalKB, a novel knowledge-based framework for predicting the binding sites of metal ions on proteins by using atomic-level statistical potentials and graph-theoretical strategies. Specifically, possible donor atom clusters are first identified using a clique detection algorithm, from which initial metal ion coordinates are generated. These candidate coordinates are then evaluated and locally refined using knowledge-based statistical potentials derived from a protein-metal ion binding database. Redundant predictions are subsequently removed by applying spatial distance thresholds. Evaluations on diverse benchmark data sets provided by Metal3D and TEMSP show that MetalKB demonstrates competitive performance compared with seven representative methods in terms of precision, recall, and F1 score, while exhibiting strong robustness and parameter stability. MetalKB is capable of identifying complex coordination environments, including multinuclear and bridging metal-binding sites, as illustrated in representative structural examples. In addition, it also provides prediction of both metal ion 3D coordinates and residue-level coordinating ligands.

1. INTRODUCTION

Metal ions play key roles in the function and stability of biomacromolecules.[1,2] Common biologically relevant metal ions include Zn^{2+}, Ca^{2+}, Mg^{2+}, Mn^{2+}, Fe^{2+}/Fe^{3+}, Cu^{2+}, Co^{2+}, Na^{+}, K^{+}, and Ni^{2+}. These ions participate in a wide range of physiological processes, such as signal transduction, electrolyte homeostasis, and enzymatic catalysis. It is estimated that approximately 30−40% of proteins require one or more metal cofactors to exert their biological functions.[3−5] In metalloproteins, metal ions typically serve structural or catalytic roles: structurally, they enhance protein rigidity through stable coordination bonds[6] or stabilize protein−protein interfaces;[7] catalytically, they lower activation energy barriers to facilitate critical biochemical reactions such as nitrogen fixation,[8] oxygenation,[9] and oxygen reduction.[10] Among them, zinc is the most abundant transition metal in the human proteome, and present in ∼10% of human proteins,[11] where it contributes both to structural stability and catalytic activity.

Given their functional importance, accurate identification of metal-binding sites on protein is crucial.[12−15,17] Experimental approaches, including mass spectrometry,[18] X-ray crystallography,[19] and surface plasmon resonance,[20] provide high accuracy in identifying metal-binding sites, but are costly and time-consuming, limiting their applicability for large-scale studies. To address the challenge, a variety of computational methods have been developed, which can be broadly categorized into sequence-based and structure-based strategies. Sequence-based methods typically identify conserved amino acid motifs associated with metal coordination. For instance, MetalDetector[21,22] focuses on cysteine- and histidine-rich regions to predict transition metal binding sites, while ZincFinder[23] employs a support vector machine trained on data sets of protein chains containing zinc-binding sites and nonmetalloproteins. However, the active sites of proteins are often composed of spatially adjacent but sequentially distant

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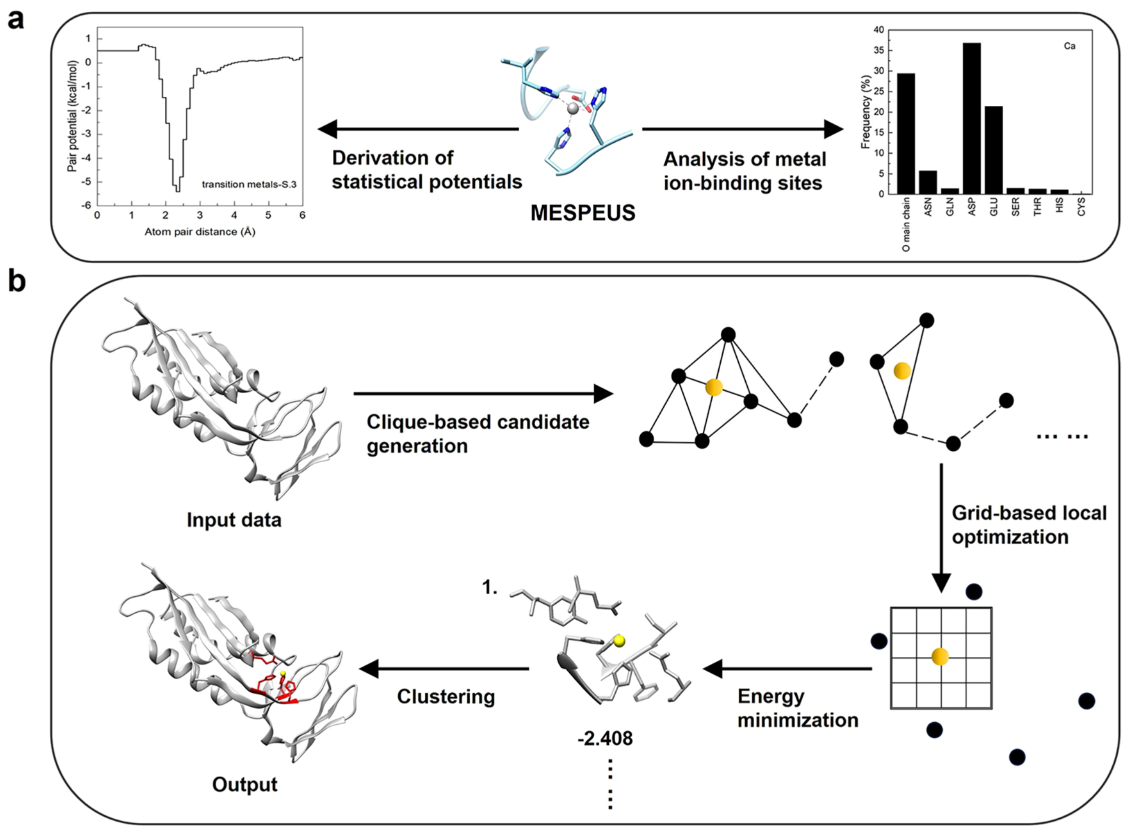

Figure 1. Workflow of MetalKB. (a) Extracting coordination geometry rules for metal ions from the metalloprotein structural data, and constructing statistical potential functions based on atom pair distributions. (b) Predicting metal ion binding sites using clique detection algorithms and statistical potential models.

residues.[24] Therefore, sequence-based approaches frequently fail to capture three-dimensional structural features and lack detailed descriptions of metal-protein atomic-level interactions.

In contrast, structure-based methods can directly utilize the three-dimensional conformational information on proteins and generally achieve higher prediction accuracy. From a methodological perspective, these approaches can be further categorized according to whether they rely on structural templates, whether they explicitly model geometric constraints, whether machine learning or deep learning models are employed, and the range of metal types they are designed to handle. Representative methods include MIB[15,25] (a template-matching approach based on structure fragment transformation,[16] which can serve as a general predictor for binding sites of various metal ions), BioMetAll[26] (a method based on backbone geometric preorganization for predicting binding sites of multiple metal ions), TEMSP[27] (a zinc-binding site prediction method based on residue-pair template matching and geometric constraints), CHED[28] (a method which identifies geometric triads in apo proteins and is mainly targeted at transition metals such as zinc), ZincBindDB[29] (a machine-learning classifier trained on zinc coordination motifs), Metal3D[30] (a three-dimensional convolutional neural network using voxelized protein structures and trained on zinc-binding site data), and PinMyMetal (PMM)[31] (a structure-based hybrid machine-learning system for predicting transition metal-binding sites).

In recent years, several newly developed methods have also been proposed, including MetalSiteHunter[32] (a multimetal binding site prediction method based on three-dimensional convolutional neural networks), ESMBind[33] (a general metal-binding site prediction method that combines protein pretrained model embeddings with physics-based energy minimization strategies), MasterOfMetals[34] (a zinc-binding site prediction method that integrates graph convolutional networks with geometric filtering based on Cα/Cβ distance matrices), and MIC[35] (a method that classifies water molecules and multiple ion types in cryo-electron microscopy and crystallographic structures using ion coordination environment information).

Despite the significant progress achieved by structure-based methods, several challenges remain. For example, template-based structural approaches often perform well when structural homologues are available, but their accuracy declines significantly for new metal-binding patterns or proteins without suitable templates. In addition, conventional geometric descriptors, such as interatomic distances and bond angles, contain limited information and are insufficient to fully represent the complex interactions between metal ions and their coordinating ligands. Quantum mechanics/molecular mechanics (QM/MM) simulations can indeed provide highly accurate descriptions of metal coordination environments;[36] however, their extremely high computational cost makes them impractical for routine large-scale prediction and screening of metal-binding sites.[37]

To address the challenges, we introduce MetalKB, a knowledge-based graph framework powered with statistical protein−ligand interaction potentials. MetalKB predicts metal ion-binding sites by combining graph-theoretic modeling with atomic-level statistical potentials. Specifically, it constructs

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Table 1. Coordination Frequencies (%) of Metal Ions with Protein Residue Donor Atoms[a]

	CA	MG	NA	K	MN	FE	CO	NI	CU	ZN
O main chain	29.39	22.51	49.20	59.94	4.28	0.27	3.55	3.03	3.22	1.38
ASN	5.74	4.97	3.61	4.14	2.42	0.49	0.72	0.93	0.16	0.39
GLN	1.41	1.62	2.18	2.28	0.95	0.26	0.48	0.74	0.43	0.22
ASP	36.82	32.10	12.82	8.03	36.82	11.42	23.60	13.61	4.77	16.96
GLU	21.43	19.48	9.56	7.10	24.76	23.62	19.57	11.82	3.08	16.50
SER	1.45	3.16	5.15	4.32	0.69	0.15	0.56	0.53	0.11	0.23
THR	1.32	4.33	4.29	4.35	0.43	0.00	0.60	0.19	0.05	0.12
HIS	1.07	8.90	1.54	1.14	27.86	51.59	44.61	53.82	60.01	31.64
CYS	0.09	0.28	0.13	0.07	0.85	5.40	3.55	7.07	21.00	31.09

[a]The first row lists different metal ions and the first column refers to protein donor atoms. Frequencies are displayed only for chemically recognized protein residue donor atoms; atoms annotated as coordinating atoms in MESPEUS based on distance criteria (such as main-chain nitrogen atoms) are excluded.

donor atom graphs under distance constraints to sample candidate metal ion positions. These candidates are then scored and refined using knowledge-based potentials, where redundant predictions are removed via spatial clustering. We benchmarked MetalKB against seven advanced methods using the data sets provided by Metal3D and TEMSP. Our results show that MetalKB achieves superior performance in terms of precision, recall, and F1-score, with strong robustness and parameter stability. In addition, MetalKB not only predicts coordinating residues but also provides 3D coordinates of metal ions, offering valuable support for downstream structural and functional analysis.

2. METHODS

An overview of the MetalKB workflow is illustrated in Figure 1. The framework consists of two main stages: (i) derivation of metal-protein statistical potentials from metalloprotein structural data, and (ii) prediction of metal-binding sites through clique-based sampling and energy-based refinement.

2.1. Data Sets

In this study, the MESPEUS[38,39] database was selected as the primary data source for analyzing metal-binding sites and deriving distance-dependent statistical potentials. MESPEUS is a curated repository dedicated to the annotation of metal ion binding sites in proteins. Since its release in 2008, it has systematically collected a large number of high-resolution metalloprotein structures from the Protein Data Bank (PDB). To ensure data quality and structural reliability, MESPEUS includes only entries determined by X-ray crystallography or cryo-electron microscopy with resolutions better than 2.5 Å, while excluding those structures resolved by nuclear magnetic resonance (NMR) or lacking resolution information. Although several alternative databases are available, MESPEUS's stringent inclusion criteria effectively reduce errors arising from low-resolution or ambiguous structures, thereby enhancing the reliability of geometric feature extraction and distance-based statistical analysis. For these reasons, MESPEUS was adopted as the primary data set in this work.

With the MESPEUS data set, we systematically analyzed the coordination preferences of metal cations in protein structures. These cations typically coordinate with specific donor atoms, including the sulfur atom from cysteine (Cys) side chains, the nitrogen atom in the imidazole ring of histidine (His), the oxygen atoms in the carboxyl groups of aspartate and glutamate (Asp/Glu), the hydroxyl oxygen atoms of threonine and serine (Thr/Ser), as well as backbone carbonyl oxygen atoms.[40,41] The frequencies of these donor residues across various metal-binding sites, as derived from MESPEUS, are summarized in Table 1. Here, the frequencies are calculated only for protein residue donor atoms and are intended to summarize the relative coordination preferences of different metal ions at the residue level.

The statistical analysis revealed that alkaline earth metals (Ca^{2+} and Mg^{2+}) exhibit a strong preference for coordinating with side-chain carboxylate oxygen atoms, followed by backbone carbonyl and side-chain hydroxyl oxygen atoms. In contrast, alkali metals (Na^+ and K^+) give a broad tendency to coordinate with all types of oxygen donors in proteins, regardless of whether they are from the backbone or side chains. Specifically, Na^+ coordinates with backbone carbonyl oxygen atoms at a frequency of 49.20%, while this frequency reaches 59.94% for K^+, indicating their weak directionality and general binding for oxygen ligands. In addition to these oxygen-dominated coordination patterns, Mg^{2+} is also observed in association with histidine residues. However, from the perspective of classical coordination chemistry, Mg^{2+} is a typical hard acid whose stable coordination behavior is dominated by interactions with oxygen donor atoms, whereas direct coordination with the imidazole nitrogen atoms of histidine does not represent a common or dominant binding mode. Consequently, Mg-His contacts should not be interpreted as an evidence for an intrinsic preference of Mg^{2+} for nitrogen donors. Moreover, part of these observations may be influenced by the difficulty in distinguishing light metal ions (such as Mg^{2+} and Na^+) from water molecules during protein structure determination, particularly in lower-resolution structures or in more flexible binding sites, where metal identity assignment may be uncertain. In comparison to Mg^{2+}, transition metals (e.g., Mn, Co, Fe, Ni, Cu, Zn) show a pronounced preference for the imidazole nitrogen atoms of histidine. Among them, Cu and Zn also exhibit a marked preference for sulfur atoms from cysteine residues.

Overall, calcium and magnesium show a distinct preference for aspartate and glutamate residues, with backbone carbonyl oxygen atoms also commonly serving as ligands for Ca, Mg, Na, and K. In contrast, transition metals such as Mn, Fe, Co, Ni, Cu, and Zn strongly prefer histidine. Mn, Fe, and Co also tend to coordinate with Asp and Glu, while Ni, Cu, and Zn show a significant preference for cysteine. Transition metals typically prefer to coordinate with four residues: histidine, cysteine, aspartate, and glutamate.

Based on the observed coordination behavior, ionic radii, and data distribution, the metal data set is classified into four categories for statistical potential derivation: (1) the group of Ca^{2+} and Na^+ ions, which share similar coordination patterns and comparable ionic radii, (2) the K^+ group, which has the largest ionic radius and distinct coordination characteristics, (3) the Mg^{2+} group, which, despite being an alkaline earth metal, possesses the smallest ionic radius and highest charge density, leading to coordination features distinct from Ca^{2+}, and (4) the Zn^{2+}-based group consisting of transition metals such as Mn, Fe, Co, Ni, Cu, and Zn, which commonly coordinate with nitrogen and sulfur donors from histidine and cysteine, as well as with carboxylate oxygen atoms. To reduce redundancy and avoid bias in the derivation of statistical potentials, we applied sequence redundancy filtering to the proteins using the CD-HIT tool with a 30% sequence identity threshold. The final statistical data set consists of 2568 nonredundant Zn-binding protein structures, 2375 Ca-binding protein structures, 3451 Mg-binding protein structures, and

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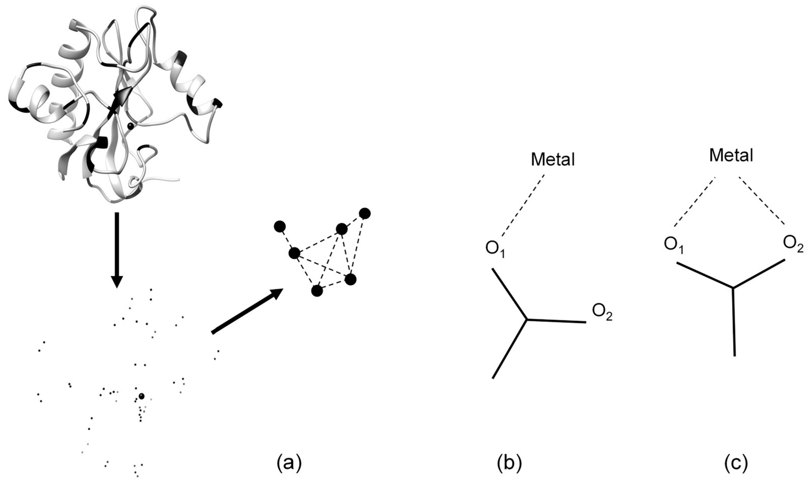

Figure 2. Clique-based candidate site sampling method in MetalKB. (a) An illustration showing how a protein structure is converted into a set of atoms and subsequently represented as a graph, where atoms correspond to nodes and edges are established only when interatomic distances fall within a coordination distance range determined from statistical analysis. (b, c) Two typical coordination modes of metal ions involving oxygen atoms from carboxyl groups.

778 K-binding protein structures. Here, metal-binding protein structures specifically refer to proteins coordinating the corresponding metal ions. It should be noted that, although the numbers of protein structures differ among metal-specific data sets, the statistical potentials in this work are derived independently for each metal type using its corresponding data set, without parameter sharing across different metals. In addition, the training data set is nonredundant with respect to the Metal3D test set and can therefore be regarded as an independent data set for performance evaluation.

2.2. Derivation of Statistical Potentials

The interaction energy between metal ions and proteins was computed by summing the pairwise potential functions $u_{ij}(r)$ over all relevant atom pairs. The total energy score is defined as follows (eq 1).

$$\text{energyscore} = \sum_{ij} u_{ij}(\text{r}) \tag{1}$$

where r denotes the distance between the metal ion i and the protein atom j. Only those protein atoms located within a radius of 6 Å from the metal ion center were considered. The potential function $u_{ij}(r)$ represents the interaction potential between a metal ion i and a protein atom of type j at distance r, and takes the weighted average of a knowledge-based potential $w_{ij}(r)$ and a van der Waals interaction potential $v_{ij}(r)$.[42−44]

The knowledge-based potential $w_{ij}(r)$ follows an inverse Boltzmann relationship with the observed interaction frequency; that is, more frequently observed interactions correspond to lower energy values. The specific expression is given as follows (eq 2).

$$w_{ij}(r) = -k_B T \log\left(\frac{\rho_{ij}^{\text{obs}}(r)}{\rho_{ij,\text{bulk}}^{\text{obs}}}\right) \tag{2}$$

where, k_B is the Boltzmann constant, T is the absolute temperature. Without loss of generality, $k_B T$ is set to be 1. This choice does not represent a true physical temperature, but rather serves to simplify the expression of the statistical potential. The term $\rho_{ij}^{\text{obs}}(r)$ represents the number density of type j protein atoms within the radial shell from $r - \Delta r/2$ to $r + \Delta r/2$ from a metal ion i. The $\rho_{ij,\text{bulk}}^{\text{obs}}$ denotes the average number density within a reference sphere of radius $R_{\max} = 10$ Å, serving as a baseline. These densities are calculated as follows (eqs 3 and 4).

$$\rho_{ij}^{\text{obs}}(r) = \frac{1}{M}\sum_{m=1}^{M} \frac{n_{ij}^{m}(r)}{4\pi r^2 \Delta r} \tag{3}$$

$$\rho_{ij,\text{bulk}}^{\text{obs}} = \frac{1}{M}\sum_{m=1}^{M} \frac{N_{ij}^{m}}{V(R_{\max})} \tag{4}$$

where M is the total number of samples, $n_{ij}^{m}(r)$ is the number of metal-protein atom pairs at distance r for the mth metal ion-binding site, N_{ij}^{m} is the total number of such pairs within the site, $V(R_{\max})$ is the volume of the reference sphere, and the shell thickness Δr is set to 0.1 Å.

The final potential function $u_{ij}(r)$ was taken from a combination of knowledge-based potential $w_{ij}(r)$ and van der Waals interaction potential $v_{ij}(r)$ as follows (eq 5).

$$u_{ij}(r) = \begin{cases} \min[w_{ij}(r), v_{ij}(r)], \; r \le 3.0 \text{ Å} \\ \dfrac{v_{ij}(r)\mathrm{e}^{-v_{ij}(r)} + w_{ij}(r)\mathrm{e}^{-w_{ij}(r)}}{\mathrm{e}^{-v_{ij}(r)} + \mathrm{e}^{-w_{ij}(r)}}, \; r > 3.0 \text{ Å} \end{cases} \tag{5}$$

Here, the van der Waals interaction $v_{ij}(r)$ is described by a Lennard-Jones 12−6 potential. Given the dominant role of short-range interactions in metal-protein coordination, the above combination approach retains conservative potential well depths at short distances by taking the minimum between the fitted potential and the van der Waals potential for $r \le 3.0$ Å, while ensuring a gradual energy decay to zero at longer ranges ($r > 3.0$ Å) through exponential weighting. This long-range behavior arises from the asymptotic properties of the two contributing terms: the knowledge-based potential $w_{ij}(r)$ approaches zero as the observed pair density converges to the bulk reference density, whereas the van der Waals term $v_{ij}(r)$ decays rapidly to zero due to its Lennard-Jones 12−6 form. In this case, the resulting score mainly reflects free-energy preferences observed in experimental structures, rather than absolute interaction energies. As a result, this hybrid method effectively integrates van der Waals interactions with statistical potentials, thereby enhancing both the robustness and physical plausibility of the overall potential energy profile.

2.3. Sampling Metal Ions through Clique-Based Identification

During metal ion sampling, a clique detection strategy was employed to extract coordination clusters with geometrically reasonable distributions, from which the spatial coordinates of the metal ions were inferred [see Figure 2a]. Here, a clique refers to a fully connected subgraph in which all nodes (atoms) are mutually connected. As illustrated in Figure 2a, the input protein structure is first reduced to a set of candidate donor atoms, which are then represented as nodes in a graph, with edges established only when interatomic distances fall within a coordination distance range determined from statistical analysis.

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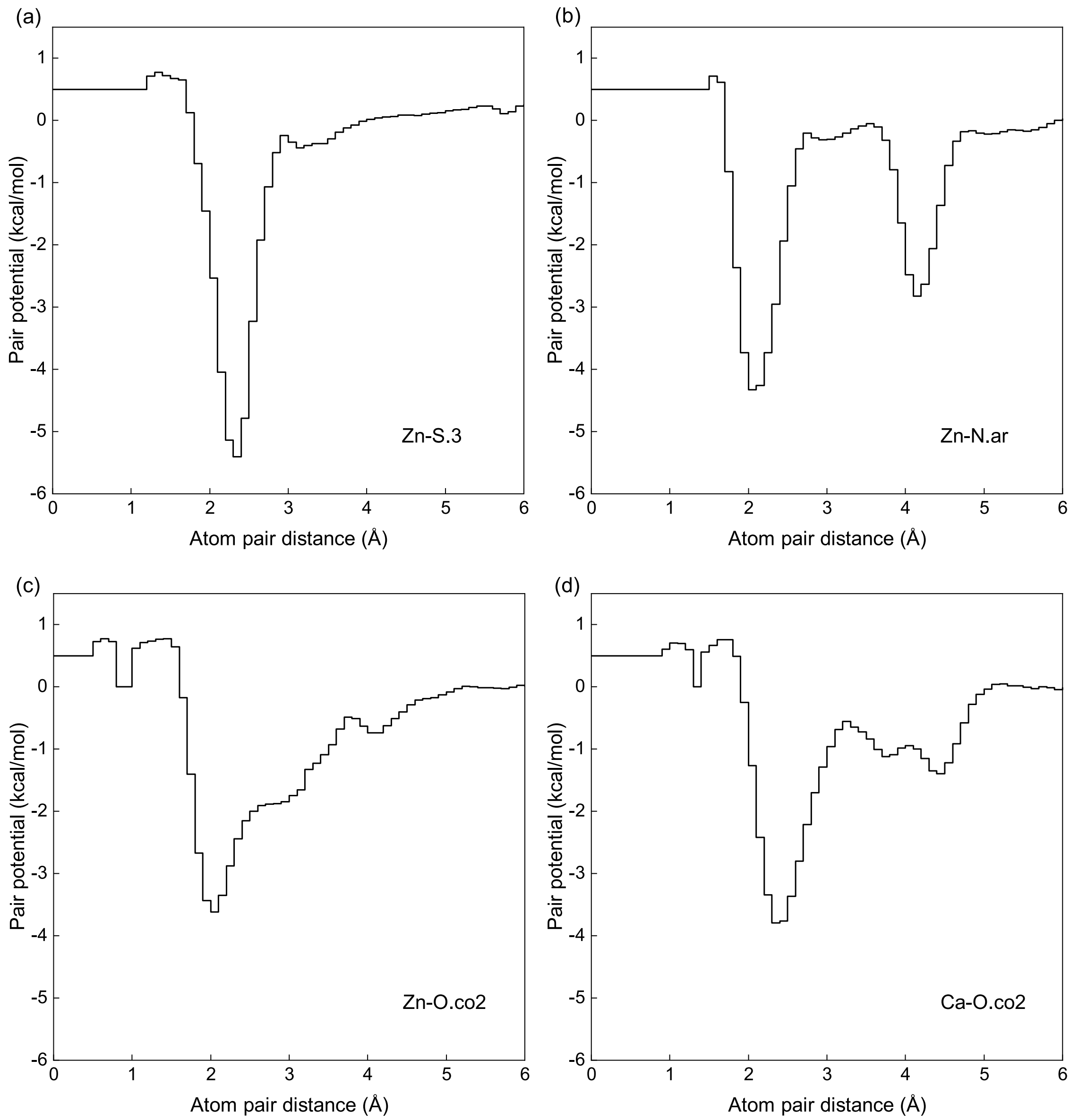

Figure 3. Four representative pair potentials selected in MetalKB. (a) Zn−S.3, (b) Zn−N.ar, (c) Zn−O.co2, and (d) Ca−O.co2. The first atom type refers to the metal ion, and the second refers to a protein atom.

The procedure consists of four steps. (1) Extraction of donor atoms: For each input protein structure in PDB format, a donor atom set was extracted. For transition metals, donor atoms include the SG atom of cysteine, the ND1 or NE2 atoms of histidine, the OE1 or OE2 atoms of glutamate, and the OD1 or OD2 atoms of aspartate. For alkaline earth and alkali metals, donor atoms comprise side-chain oxygen atoms of glutamate, aspartate, asparagine, glutamine, serine, and threonine, as well as backbone oxygen atoms of all residues. Based on pairwise distances between these donor atoms, a graph $G(V, E)$ was constructed, where V is the set of atoms and E is the set of edges. For transition metals, an edge was added between two atoms if their distance falls within 2.4−5.2 Å, a range determined from the statistical distribution of donor−donor distances observed in the data set (Supporting Information Figure S1). (2) Clique identification. Within graph G, cliques Q were identified under the following conditions: all vertex pairs in the clique must be fully connected, the clique size must be at least three (to meet the coordination requirement of at least three donors in realistic metal sites), and cliques are allowed to share vertices, as different sites may involve the same coordinating residue. Each clique was treated as a potential coordination cluster. (3) Redundant clique removal. Some cliques may be fully included by larger ones. To reduce computational redundancy and avoid repeated predictions, smaller cliques that are strict subsets of larger ones were removed. Specifically, if Q2 is entirely contained in Q1, and Q1 contains more vertices, only Q1 was retained. (4) Estimation of metal ion coordinates. For each retained clique, the metal ion coordinate was estimated as the geometric centroid of the donor atoms (eq 6)

$$x = \frac{1}{n}\sum_{i=1}^{n} x_i,\ y = \frac{1}{n}\sum_{i=1}^{n} y_i,\ z = \frac{1}{n}\sum_{i=1}^{n} z_i \tag{6}$$

where n is the number of donor atoms in the clique, and (x_i, y_i, z_i) represents the 3D coordinate of the ith atom.

Although grid scanning can also be used to sample potential metal ion positions, this strategy has several limitations. Uniform sampling across the entire protein structure generates a large number of grid points in nonbinding regions, increasing computational overhead. More critically, it fails to accurately capture the spatial relationships among donor atoms, particularly when a single residue contributes multiple donors. For example, the carboxylate groups of glutamate and aspartate provide two oxygen atoms that may participate in diverse coordination modes, such as monodentate, bidentate, asymmetric, or bridging coordination. As shown in Figure 2b, grid-

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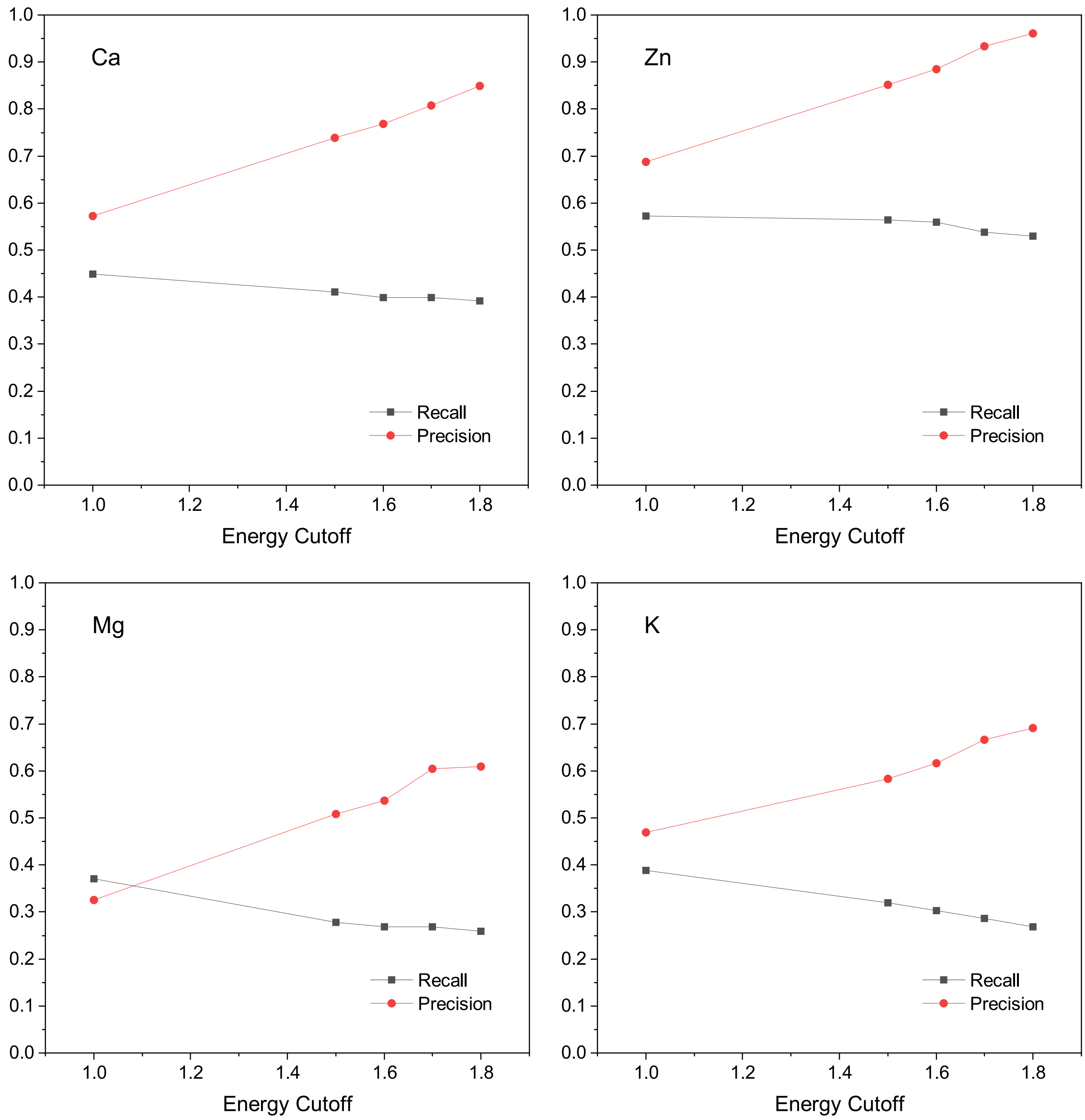

Figure 4. Precision and recall trends of MetalKB on randomly selected data set at different parameter thresholds. The *x*-axis represents different energy cutoff values, and the *y*-axis shows the corresponding values. The energy cutoff values shown are the absolute values of the total energy scores after translation and scaling.

based sampling struggles to distinguish these modes effectively. In contrast, our graph construction here allows clear differentiation of distinct modes. Specifically, during graph construction, we introduce virtual donor atom nodes between each pair of carboxylate oxygen atoms to represent potential bidentate coordination. Strict connection rules were enforced: no edge was allowed between the two oxygen atoms of the same carboxyl group, nor between either oxygen atom and its associated virtual node. This design partitions each carboxyl group into three mutually exclusive units, thereby improving the representation of diverse coordination geometries.

In addition, unlike distance-based spatial clustering, our redundancy removal strategy relies on inclusion relationships among donor clusters. This approach explicitly accounts for potential residue sharing across coordination sites and preserves multiple distinct binding patterns, thus avoiding erroneous elimination of biologically meaningful binding sites due to mere spatial proximity.

2.4. Energy Scoring and Site Clustering

Following metal ion sampling, MetalKB employs the previously derived statistical potential to evaluate the energies of all candidate binding sites. The basic principle at this stage is to identify the binding sites that best resemble native coordination environments by minimizing the potential energy.

In many cases, metal ions are relatively far apart in space, but in some instances, they are connected by chemical bridges and can be very close to each other. According to the literature,[40,45] the distance between most metal ions in multinuclear clusters typically falls within the range of 3 to 4 Å. Therefore, when two predicted metal ions are found to be significantly closer than this range, one of them should be considered as a redundant prediction and removed. To improve sensitivity in detecting multinuclear sites, the redundancy threshold was set to 2.5 Å in this study.

To enhance the precision of metal ion coordinate predictions, we introduced a local grid-based refinement strategy. Specifically, for each sampled metal ion position, a set of fine-tuned candidate points was generated within a 2.5 Å radius, with a step size of 0.25 Å. Each candidate coordinate was scored using the statistical potential function, and the coordinate with the lowest energy was selected as the final predicted position. This fine-grained scanning enables MetalKB to more accurately locate the global energy minimum for metal binding, thereby improving both the physical plausibility and spatial accuracy of the predictions.

However, this optimization process may produce predicted sites that are too close to each other, resulting in redundancy. To further eliminate such redundancies, we implemented a merging algorithm based on energy score ranking. Specifically, all optimized predicted sites were first ranked in ascending order of energy. Then, the pairwise distances between metal ions were examined. If any two predicted ions were found to be closer than 2.5 Å, only the one with the lower energy score was retained.

In the final output, based on our statistical findings, only donor residues located within 4 Å of the predicted metal ion coordinates are reported. For example, in the case of zinc, only cysteine, histidine, glutamate, and aspartate residues are included in the output.

3. RESULTS AND DISCUSSION

3.1. Derived Pairwise Potentials

Figure 3 illustrates the potential energy curves for four representative interactions between metal ions and coordinating protein atoms. These include Zn with the sulfur atom of cysteine (S.3), the nitrogen atom of imidazole (N.ar), and the oxygen atom of carboxylate (O.co2), as well as Ca with the carboxylate oxygen atom (O.co2). As shown in the figure, the Zn−S.3 interaction reaches its energy minimum at approximately 2.3 Å, while both Zn−N.ar, and the Zn−O.co2 interactions show minima near 2.0 Å. These distances agree well with the typical coordination bond lengths observed in most transition metals,[45] suggesting that the coordination behavior of Zn is generalizable to other transition metal ions. For Ca, the Ca−O.co2 interaction exhibits a minimum around 2.3 Å, closely matching experimentally observed distances and thereby further validating the physical reasonableness of the derived potential functions.

As shown in Figure 3a, the Zn−S.3 curve exhibits a deep and steep energy well near 2.3 Å, indicating the formation of a strong and rigid bond between Zn and the cysteine sulfur atom. This feature is consistent with the structural role of Zn-sulfur coordination in biological systems. Due to the maximal charge transfer from Cys^- to Zn^{2+}, the Lewis acidity of Zn is significantly reduced, while steric repulsion among the bulky sulfur atoms further prevents Zn^{2+} from coordinating with additional ligands such as water molecules or substrates.[46,47] A typical example is the zinc finger domain, which contributes to protein folding and stability.

Figure 3b shows that the Zn−N.ar interaction possesses two potential wells, located at approximately 2.0 and 4.1 Å. The first well corresponds to a strong coordination bond between Zn and a nitrogen atom (either $N\delta$ or $N\varepsilon$) within the imidazole ring of histidine. The second, shallower well at 4.1 Å arises from steric hindrance effects. In contrast, the Zn−O.co2 curve [Figure 3c] presents a relatively broad and shallow well, ranging from approximately 2.0 to 4.0 Å. This feature results from the ability of both oxygen atoms in the carboxylate group to coordinate with the metal ion either separately or simultaneously. Consequently, various coordination modes may form, including monodentate, symmetric bidentate, asymmetric bidentate, and bridging coordination. In monodentate coordination, a single oxygen atom forms a relatively strong coordination bond with the zinc ion (with a bond length of approximately 2.0−2.2 Å), whereas in bidentate coordination, both oxygen atoms can participate in coordination through a dynamic equilibrium mechanism.

Comparative analysis of the potential energy curves for the three Zn coordination modes reveals that the Zn−S.3 interaction features the deepest and narrowest energy well, consistent with its role in maintaining structural stability. In contrast, the Zn−N.ar and Zn−O.co2 interactions exhibit broader and shallower wells, an energy characteristic that aligns with their roles at catalytic active sites. The potential energy curve of Ca−O.co2 reaches its minimum within the range of 2.2−2.4 Å, with a secondary energy well at approximately 4.5 Å. This reflects the coordination preference of Ca: unlike Zn, which exhibits a continuous range between monodentate and bidentate coordination, Ca tends to favor either monodentate or symmetric bidentate coordination (where the two Ca−O bond lengths are nearly identical).

Based on the energy threshold, the confidence of predicted binding sites can be assessed, with the energy magnitude serving as a relative ranking indicator. For ease of use, we applied translation and scaling to the total energy and recommend setting the energy threshold at 1.7 (the original energy scores are negative, and for convenience only their absolute values are reported in subsequent analyses). Figure 4 shows the prediction results for 100 randomly selected structures from the Ca, Zn, Mg, and K statistical data sets. As shown from the figure, with increasing energy cutoff values, precision consistently improves, while recall shows an opposite trend, reflecting a typical precision-recall trade-off. Accordingly, an energy threshold of 1.7 was chosen as a compromise, providing substantially improved precision while maintaining recall at an acceptable level across different metal types. It should be noted that our research focuses on the interaction between metal ions and proteins, so small molecules were not identified during the derivation of the statistical potentials, nor were cases with a metal-protein coordination number less than three specifically handled.

It should be noted that the metal type must be specified as an input in our method, since MetalKB employs metal-specific statistical potentials that are independently derived for each metal category. An interesting question is whether the resulting energy scores can be further used to discriminate metal types at a given binding site. A preliminary analysis addressing this question is provided in the Supporting Information (Figure S2). It is revealed that the statistical potentials exhibit a certain degree of metal-type specificity. However, because MetalKB derives its statistical potentials based on representative metals for each category and different metal-binding sites may exhibit similarities in their geometric and coordination features, further work is required to enable more detailed discrimination among specific metal types.

3.2. Assessment on Metal3D Test Set

To evaluate the performance of MetalKB in predicting metal ion binding sites, we conducted a comparative analysis using the test set provided by Metal3D.[30] Since an energy threshold of x is normally needed during the evaluations, we define "MetalKB(x)" as the prediction results for the energy threshold of x.

The Metal3D test set for zinc ion binding sites includes 59 proteins, comprising a total of 189 sites. To remove the redundancy, we manually curated the data set following the strategy outlined in the PMM[31] paper, ultimately retaining 178 zinc-binding sites to align with prior studies. Our evaluation criteria followed the standards defined in the Metal3D paper: a prediction is considered a true positive (TP) if the distance to the true metal ion coordinate is ≤5 Å. Multiple predictions for the same site count as a single TP. A true metal-binding site is considered a false negative (FN) if it is not covered by any

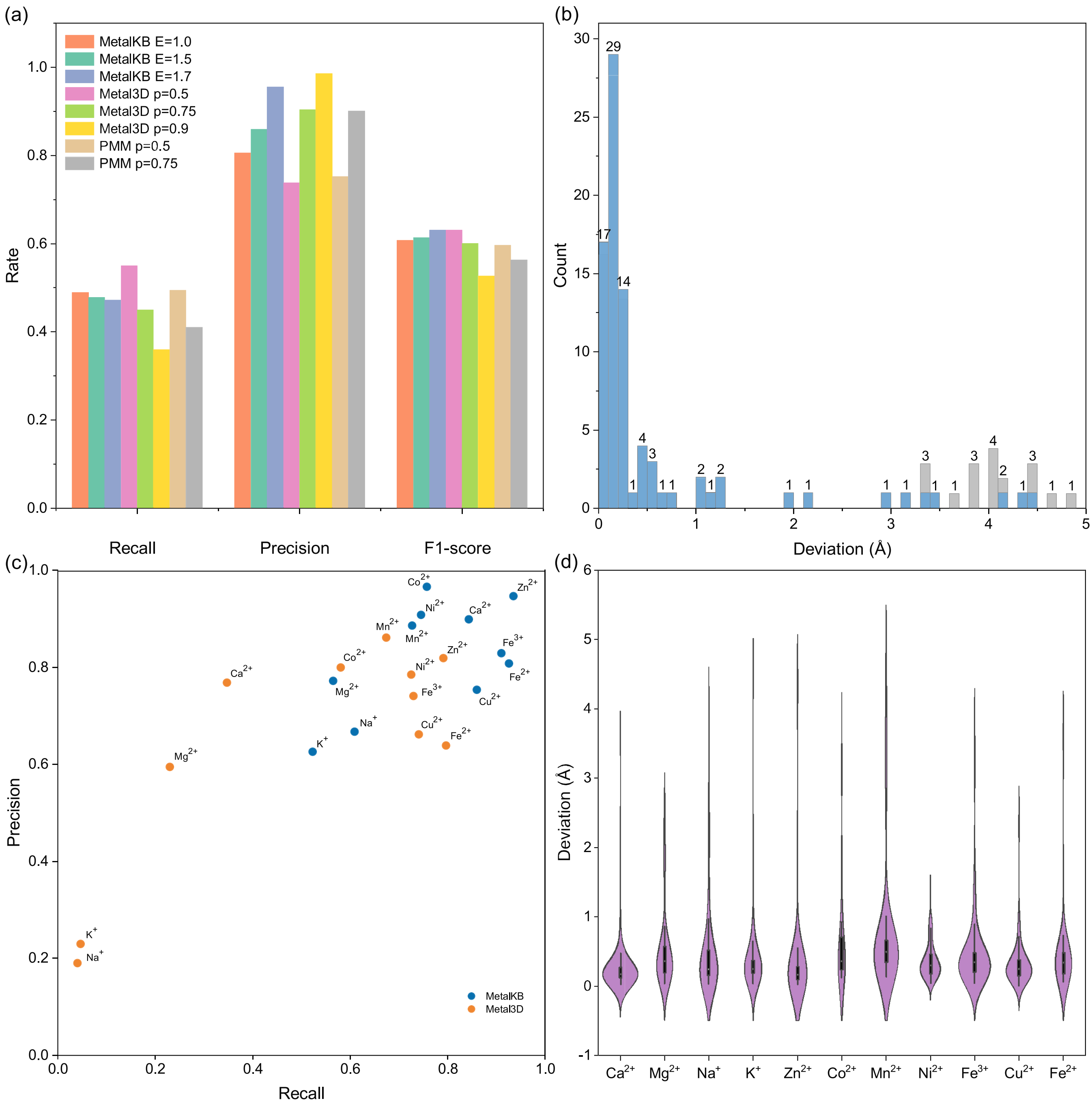

Figure 5. Performance of MetalKB on the Metal3D test set. (a) Comparison of three methods (MetalKB, Metal3D, and PMM). (b) Distance distribution of MetalKB predictions, where gray bars represent the predictions affected by multinuclear metal sites. (c) Comparison between MetalKB (blue, energy threshold = 1.7) and Metal3D (orange, p = 0.75) for 11 types of metal-binding sites in the test set. (d) Distance distribution of MetalKB predictions for the 11 types of metal-binding sites. Negative values reflect signed deviations relative to the reference position and do not represent physically negative distances.

prediction within 5 Å. A prediction is considered a false positive (FP) if it is more than 5 Å from any true binding site and if the distance between all predictions exceeds 5 Å. Precision, recall, and F1-score were calculated based on these criteria. Precision is defined as the fraction of predicted binding sites that are true positives, recall is defined as the fraction of true binding sites that are correctly predicted, and the F1-score is the harmonic mean of precision and recall.

Figure 5a illustrates the performance of MetalKB, Metal3D, and PMM under various thresholds. The results show that MetalKB maintains high stability and robustness across different energy thresholds. At an energy threshold of 1.0, the precision is 0.806, the recall is 0.489, and the F1-score is 0.608. Tightening the threshold to 1.5 increases the precision to 0.859, with the F1-score rising to 0.614. At the energy threshold of 1.7, the precision increases to 0.955, although the recall slightly decreased to 0.472, while the F1-score improves to 0.631. This trend indicates that lowering the energy threshold effectively reduces false positives while maintaining a relatively stable recall, demonstrating the effectiveness of the

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energy-based scoring strategy in identifying true metal binding sites.

When compared with other methods, PMM showed a recall of 0.494 at $p = 0.5$, which is similar to MetalKB (1.0), but with a lower precision of 0.752. Raising p to 0.75 increases the precision to 0.901, but the recall dropped to 0.410, and the F1-score decreases to 0.563, indicating that the method's sensitivity to parameter changes. Metal3D achieves an F1-score of 0.631 at $p = 0.5$, which is comparable to MetalKB (1.7). However, when p is set to 0.75 and 0.9, the precisions increase to 0.904 and 0.986, but the recall drop to 0.450 and 0.360, leading to significantly lower F1-scores of 0.601 and 0.527, respectively, reflecting the high sensitivity of Metal3D to parameter settings.

Regarding spatial prediction accuracy, MetalKB (1.7) shows an average error of 1.117 Å, with a standard deviation of 1.567 Å, which is higher than that of Metal3D for $p = 0.75$, which has an error of 0.710 ± 0.631 Å. However, MetalKB has a small median error of 0.224 Å, outperforming the 0.508 Å of Metal3D. This discrepancy may be understood because Metal3D uses a 5 Å clustering threshold, which adequately covers the spatial distribution of multinuclear zinc ions and may impact error assessment. To further investigate the robustness, we analyzed the prediction error distribution of MetalKB [Figure 5b]. It is found that 15 predictions with errors greater than 3 Å are due to the identification of dinuclear sites (indicated by the gray bars). After excluding these predictions, the average error of MetalKB reduces to 0.596 Å, with a standard deviation of 1.025 Å.

For a more comprehensive evaluation of the performance across different metal types, we evaluated MetalKB on the multimetal prediction test set developed by Metal3D. The test set includes 11 biologically relevant metal ions: Ca^{2+} (100), Mg^{2+} (100), Na^{+} (100), K^{+} (100), Mn^{2+} (100), Fe^{3+} (100), Fe^{2+} (57), Co^{2+} (30), Ni^{2+} (93), Cu^{2+} (68), and Zn^{2+} (59). These metal-binding sites are associated with at least three unique protein ligands, with an occupancy greater than 0.5. For the performance comparison across different transition metals shown in Figure 5c,d, MetalKB uses the same energy threshold of 1.7 for all metal types, whereas the Metal3D results were taken from its original publication, corresponding to a parameter value of $p = 0.75$.

Figure 5c presents the performance of MetalKB (1.7) and Metal3D ($p = 0.75$) on this test set. The results indicate that MetalKB outperforms Metal3D across most metal types, particularly excelling with Zn^{2+}, Ca^{2+}, and Fe^{3+}. Metal3D performs worse on nontransition metals such as Na^{+}, K^{+}, and Mg^{2+}. This can be attributed to its training set that is mainly limited to zinc ions, resulting in poor recognition of alkali and alkaline earth metals.

We also calculated the minimum distance between predicted points and true positive sites for each metal type [Figure 5d]. The results show that the median prediction error for MetalKB is approximately 0.3 Å, indicating that half of the predictions are within or better than 0.3 Å of accuracy. Detailed numerical values of the distance errors and comparisons with Metal3D and PMM for transition metals are provided in the Supporting Information (Table S1).

3.3. Assessment on TEMSP Test Set

Given the importance and unique functions of zinc in proteins, accurately identifying zinc-binding sites is of great significance. To further evaluate the performance of the MetalKB method, we tested it on the TEMSP[27] benchmark data set, which consists of 100 protein structures encompassing 136 experimentally validated zinc-binding sites. Unlike the distance-based matching criterion used in the Metal3D benchmark, the TEMSP data set adopts the Intersection over Union of Residues (IoUR) as the evaluation metric for prediction performance. The IoUR is calculated as follows (eq 7)

$$\text{IoUR} = \frac{N(\text{Predicted ligand residues} \cap \text{True ligand residues})}{N(\text{Predicted ligand residues} \cup \text{True ligand residues})} \tag{7}$$

An IoUR value of 1 indicates the perfect overlap between the predicted and true ligand residues. According to the original TEMSP publication, a predicted site is considered a true positive (TP) if its IoUR is ≥ 0.5, indicating that at least half of the ligand residues are correctly predicted. Predictions with IoUR < 0.5 are considered false positives (FP). Any experimentally verified binding sites that are not captured by predictions are treated as false negatives (FN). Compared with distance-based criteria, the IoUR metric is more stringent. It not only requires spatial proximity between the predicted and true sites but also demands high consistency in the composition of ligand residues.

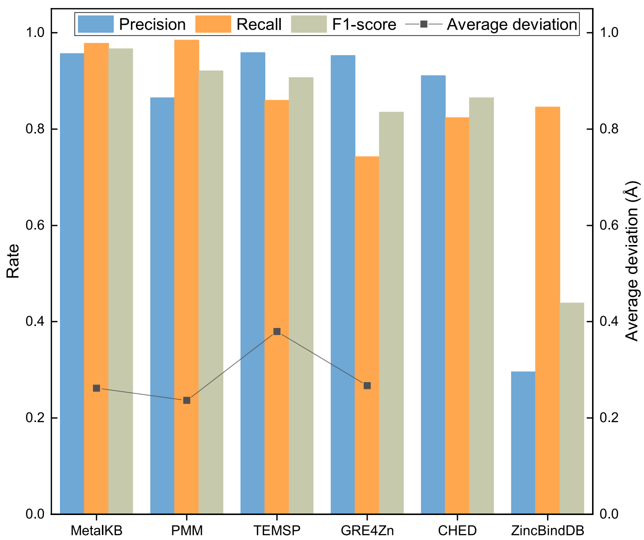

Figure 6. Performance comparison of six methods (MetalKB, PMM, TEMSP, GRE4Zn, CHED, ZincBindDB) for zinc-binding sites on the TEMSP benchmark data set. The bars show the precision, recall, and F1-score, while the line displays the average distance deviation (Å). Average coordinate deviations are not reported for CHED and ZincBindDB because these methods predict binding sites only and do not provide explicit three-dimensional coordinates of metal ions.

As shown in Figure 6 and Table 2, the MetalKB's recall rate is 0.978, which is slightly lower than PMM's 0.985, but it significantly outperforms GRE4Zn (0.743), TEMSP (0.860), ZincBindDB (0.846), and CHED (0.824). In addition, MetalKB achieves a precision of 0.957 with only 6 false positives, which is superior to PMM's 0.865. MetalKB's F1 score reaches 0.967 (the highest among all methods), indicating a well-balanced performance between precision and recall. In contrast, although GRE4Zn and TEMSP exhibit

Table 2. Performance Comparison of MetalKB with Five Other Zinc-Binding Site Prediction Methods (PMM, GRE4Zn, TEMSP, ZincBindDB, CHED) on the TEMSP Benchmark[a]

method	TP	FN	FP	precision	recall	F1 score	deviation (Å)
MetalKB	133	3	6	0.957	0.978	0.967	0.262
PMM	134	2	21	0.865	0.985	0.921	0.237
TEMSP	117	19	5	0.959	0.860	0.907	0.380
CHED	112	24	11	0.911	0.824	0.865	
GRE4Zn	101	35	5	0.953	0.743	0.835	0.267
ZincBindDB	115	21	273	0.296	0.846	0.439	

[a]The methods are sorted according to their F1 Scores. The data for the methods other than MetalKB were obtained from the literature.[31]

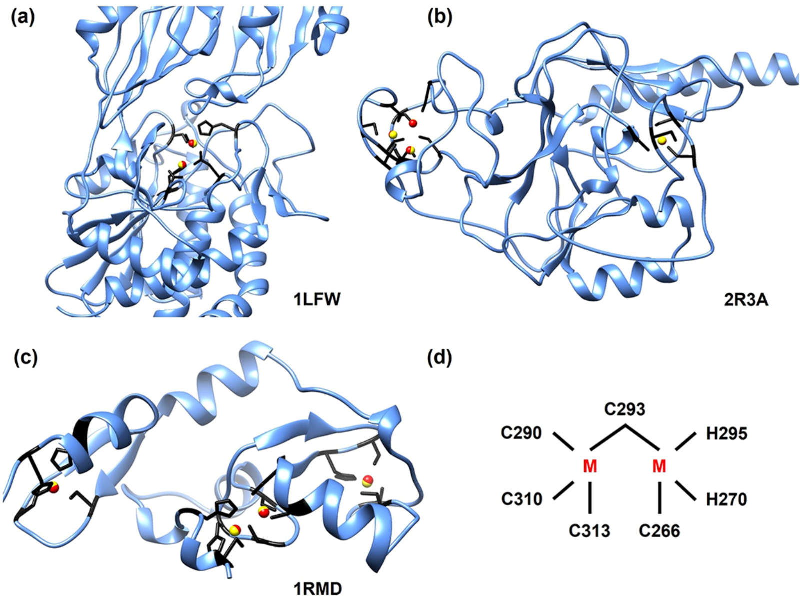

Figure 7. Examples of MetalKB predictions for multinuclear zinc-binding sites. (a) Dinuclear zinc aminopeptidase PepV from Lactobacillus. (b) Human H3K9 methyltransferase. (c) RAG1 dimerization domain. (d) Dinuclear zinc cluster in the RAG1 dimerization domain. In the figures, gold spheres represent the metal ion positions from experimentally resolved structures, while red spheres indicate the metal ion positions predicted by MetalKB.

an excellent precision (0.953 and 0.959, respectively), their recall is notably lower, particularly for GRE4Zn, which has a recall of only 0.743, indicating a high risk of missing correct predictions. The issue of false positives is most prominent in ZincBindDB, which records as many as 273 false positives, resulting in a low precision of 0.296 and an F1 score of just 0.439. In terms of coordinate prediction accuracy, all methods perform well. For MetalKB, the average coordinate deviation is 0.262 Å, which is slightly higher than 0.237 Å for PMM and comparable to 0.267 Å for GRE4Zn.

The differences in precision and recall observed across Figures 4–6 mainly come from the differences in the composition of tested data sets, particularly the inclusion of metal-binding sites with fewer than three coordinating atoms in the data sets used for Figures 4 and 5a, compared with the more canonical coordination environments represented in Figures 5c and 6.

3.4. Examples for Multi-Nuclear Zinc Site Prediction

To further evaluate the capability of MetalKB in recognizing complex metal-binding environments, we assessed its performance on multinuclear metal site prediction tasks. Figure 7 presents the visualization of representative structures, where gold spheres indicate experimentally determined zinc ion positions and red spheres represent those predicted by MetalKB.

Figure 7a shows the typical dinuclear zinc peptidase PepV from Lactobacillus, with its catalytic center containing two zinc ions (Zn1 and Zn2).[48] Zn2 is coordinated by His87, Asp119, and Asp177, while Zn1 is coordinated by His439, Asp119, and Glu154. The Asp119 acts as a bridging ligand, forming a bimetallic coordination architecture with an intermetallic distance of approximately 3.8 Å. The Zn1 facilitates substrate binding and, together with His269-NE2, polarizes the scissile carbonyl oxygen, thereby promoting nucleophilic attack by the catalytic water molecule. The Zn2 activates the water molecule and facilitates binding and hydrolysis. These cooperative roles ensure the efficiency and specificity of the reaction. MetalKB successfully identifies and precisely predicts the positions of both Zn1 and Zn2, with coordinates highly consistent with experimental data. The average intermetallic distance error is less than 0.18 Å, and all key coordinating residues are correctly identified. In particular, the accurate recognition of the bridging residue Asp119 highlights MetalKB's strong discriminative capability in handling multicenter systems with shared ligands.

In the crystal structure of human H3K9 histone lysine methyltransferase[49] [Figure 7b], the zinc ions are present in both the Pre-SET and Post-SET regions. In the Pre-SET region, three zinc ions are coordinated by nine conserved cysteine residues, forming a triangular zinc cluster. This cluster

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is buried and plays a structural role.[50] In the Post-SET region, the zinc ion is coordinated tetrahedrally with cysteine residues, and this region may be involved in SAM binding.[51] MetalKB accurately locates all zinc-binding sites and successfully recapitulates the coordination patterns and geometric configurations in both regions, showing high consistency with the crystal structures.

In the crystal structure of the RAG1 dimerization domain[52] [Figure 7c], the zinc ions participate in three distinct coordination environments: a canonical mononuclear C3H-type RING finger, a C2H2-type zinc finger, and a unique $Zn_2Cys_5His_2$ binuclear zinc cluster. In the latter, Cys293 serves as a bridging ligand coordinating both zinc ions, while other ligands include Cys266, His270, and His295. These zinc sites play crucial roles in the proper folding, structural stability, and dimer interface formation of the RAG1 dimerization domain. MetalKB successfully identifies all zinc-binding sites, and in particular, captures the spatial relationship, and shares ligand features of the binuclear cluster with high precision, demonstrating its robust capability in recognizing atypical, multicenter, and geometrically complex metal-binding environments. Although this section primarily presents the predictions of zinc-binding sites, several representative examples involving other metal types are included in the Supporting Information for completeness (Figure S3).

4. CONCLUSIONS

Metal ions play critical roles in maintaining protein structural stability and regulating biological functions. Accurate identification of their binding sites remains a fundamental challenge in structural biology. In this work, we present MetalKB, a knowledge-based framework that combines graph-theoretical modeling with atom-level statistical potential functions for the prediction of metal ion-binding sites on proteins. We constructed metal-specific potential functions based on the MESPEUS database and implemented a hybrid strategy integrating cluster-based identification and local grid-based refinement for binding site prediction. Evaluations on the Metal3D and TEMSP benchmark data sets demonstrate that MetalKB shows a stable and reliable predictive performance across different metal categories and test scenarios.

MetalKB is constructed based on explicitly derived, metal-specific statistical potentials, and its prediction process can be directly associated with atom-level interactions, thereby providing a clear physical basis for interpreting the prediction results. In addition, MetalKB supports diverse types of metal ions and outputs the predicted 3D coordinates of metal ions together with their coordinating residues. These features make MetalKB a valuable tool for mechanistic studies involving metal-protein interactions.

■ ASSOCIATED CONTENT

Data Availability Statement

The MetalKB software package can be accessed at https://github.com/huang-laboratory/MetalKB/ or http://huanglab.phys.hust.edu.cn/MetalKB/. The source code and executable program have also been archived in the Zenodo repository at 10.5281/zenodo.18999183. The other data sets used for statistical potential derivation and for testing MetalKB are available in the Supporting Information or from the Protein Data Bank.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.6c00453.

Coordinate deviation comparison (Table S1); statistical distribution of donor−donor distances (Figure S1); metal-type discrimination analysis (Figure S2); additional prediction examples (Figure S3) (PDF)

Metal3D zinc-binding test set (Table S2); Metal3D multimetal test set (Table S3); TEMSP zinc-binding test set (Table S4); data sets used for statistical potential derivation: Zn (Table S5), Mg (Table S6), Ca (Table S7), and K (Table S8) (XLSX)

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Notes

The authors declare no competing financial interest.

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